

EUII with Dual Criterion and Bayesian Design

Saverio Fontana

March 19, 2026

1 EUII implementation examples

The following 4 examples are taken from the paper from Gsponer [1].

2 Example 3.1: Design of a proof-of-concept trial in Crohn's disease

The trial evaluates a dual success criterion requiring strong evidence of a positive treatment effect

$$p(\delta > 0 | \text{Data}) \geq 0.95$$

and evidence of a clinically relevant effect size

$$p(\delta > 50 | \text{Data}) \geq 0.50$$

The design also incorporates a futility criterion defined as

$$p(\delta < 40 | \text{Data}) \geq 0.90$$

In the first stage, 10 patients are allocated to placebo and 20 to the experimental treatment. If neither the success nor the futility criteria are met at the interim analysis, a second stage enrolls an additional 10 placebo and 20 experimental patients. At the final analysis, both success and futility criteria are evaluated again, allowing for an indeterminate outcome if neither bound is crossed.

The design uses historical data via a meta-analytic-predictive approach, resulting in an informative prior on the placebo group equivalent to $n_{\text{virtual}} = 20$ patients. Denote as n_{virtual} the virtual sample size due to the prior

Basically we inspect 4 variants of the same experiment:

- sequential design with 2 groups with 10 placebo and 20 treatments per group with a $n_{\text{virtual}} = 20$
- sequential design with 2 groups with 20 placebo and 20 treatments per group and uninformative prior
- fixed design with 60 patients and $n_{\text{virtual}} = 20$
- fixed design with 80 patients and uninformative prior

The main caveat here is the perfect alignment of the prior with the true control value. We are giving the informative designs perfect historical information, and this reflect in an EUII which is strictly larger for the two informative designs with respect to the uninformative ones.

2.1 Results

In Table 1 we can see the sample size for the two sequential designs:

NOTE: in these plots and for the rest of the file, if not specified, the prior for the alternative used in the EUII computations is 10%, i.e. $P(H_1) = 0.1$

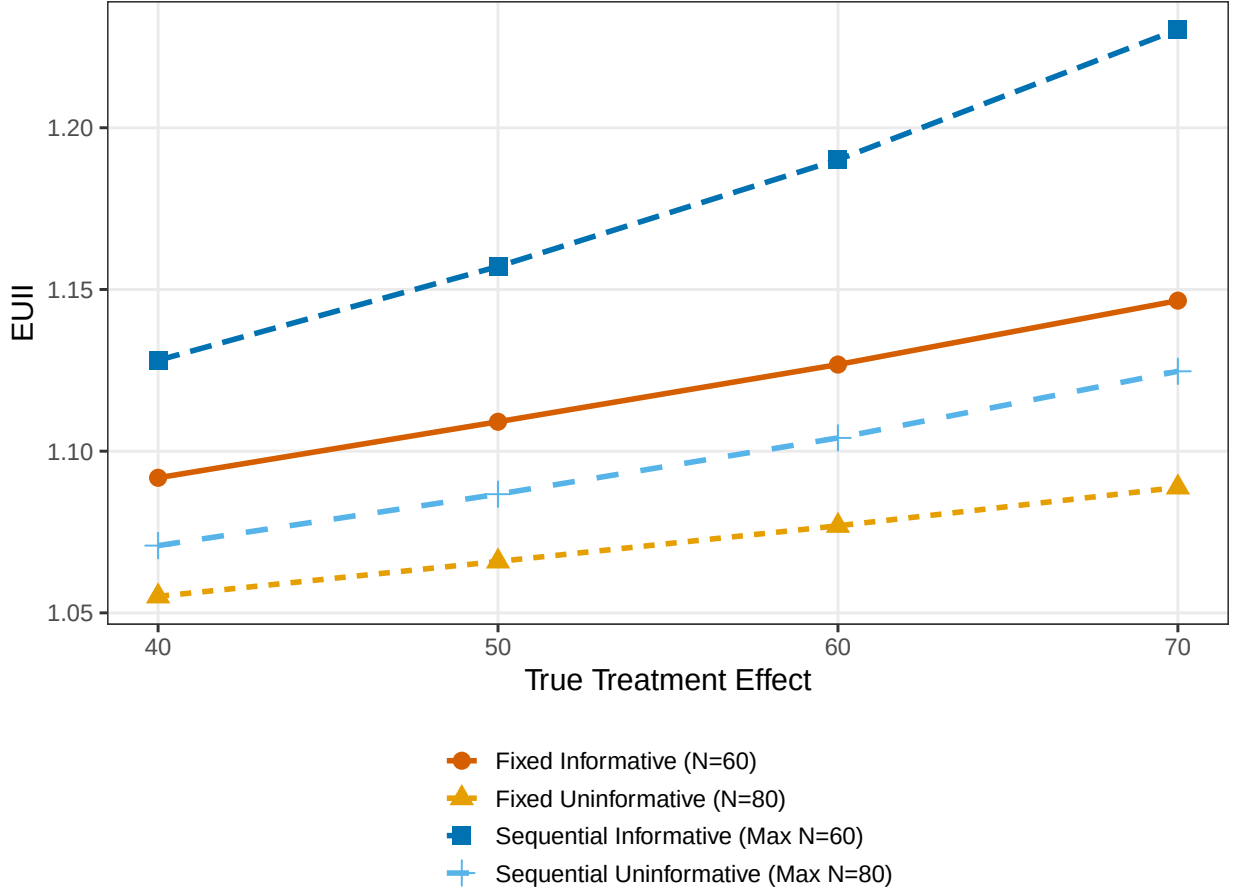


Figure 1: EUII Comparison

2.2 Sensitive analysis of varying the true value

The idea here is to inspect what happens to the EUII when there is a mistake in specifying the prior. For this purpose, we fix the prior value for control: $\mu_{C,\text{prior}} = 49$, and we inspect the resulting EUII when the real control value ($\mu_{C,\text{true}}$) varies. We define the prior bias:

$$b = \mu_{C,\text{true}} - \mu_{C,\text{prior}}$$

Note that we fix the true treatment effect $\Delta = 50$ and let b vary. It's trivial to notice that the uninformative designs are not affected by varying b .

The pattern is clear, as b increases from -20 to 20, both T1E and Power increases. Regarding the EUII, there is a slight decreasing trend, but the bumps are due to simulations. Infact, when $b < 0$ (i.e. when $\alpha \approx 0$), the resulting EUII can vary. But in all seeds I tried, the decreasing trend was always present (not the bumps). Brief intuitive explanation I thought of:

- Negative Bias ($b < 0$): The prior overestimates the true control mean. So the posterior estimate for the control is pushed higher than the data suggests. Since we don't have a prior on treatment, the resulting estimated treatment effect ($\hat{\Delta} = \text{Treatment} - \text{Control}$) is reduced. Under H_0 , the model perceives the treatment as harmful, driving the Type-I Error (α) to near zero. Under H_1 , it becomes harder claim significance, so the Power reduces.
- Positive Bias ($b > 0$): The prior underestimates the true control mean. So the posterior estimate for the control group is pushed lower than reality. This inflates the perceived treatment effect ($\hat{\Delta}$).

Δ	$E[N_+]^{\text{info}}$	$E[N_-]^{\text{info}}$	$E[N_+]^{\text{uninfo}}$	$E[N_-]^{\text{uninfo}}$
40.00	35.66	41.73	45.37	57.60
50.00	35.55	41.44	45.84	57.38
60.00	34.62	41.13	45.67	57.11
70.00	33.25	40.90	44.76	56.88

Table 1: Sample size for sequential designs

Consequently, obtaining a significant result becomes much easier. This increases statistical Power, but also T1E.

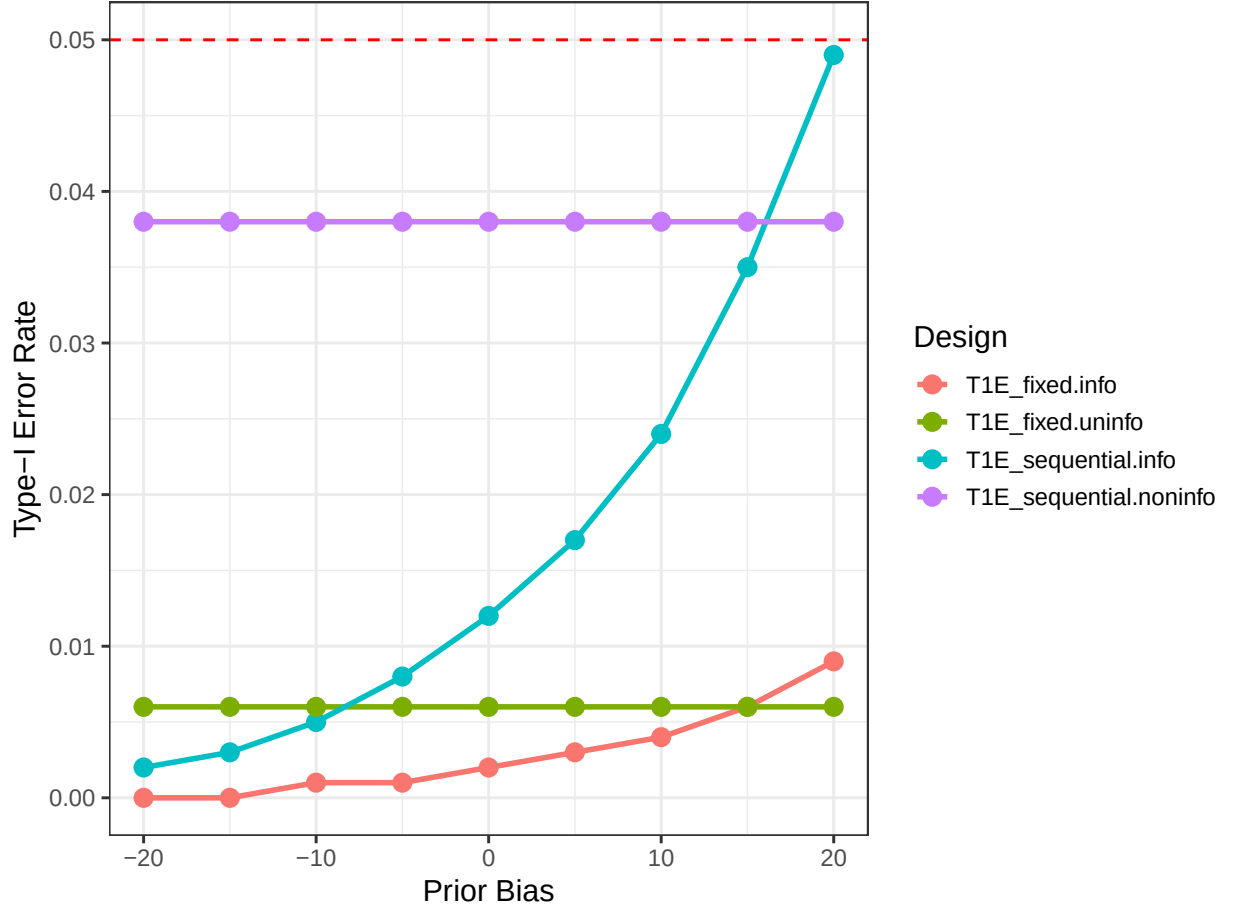


Figure 2: T1E with varying prior bias

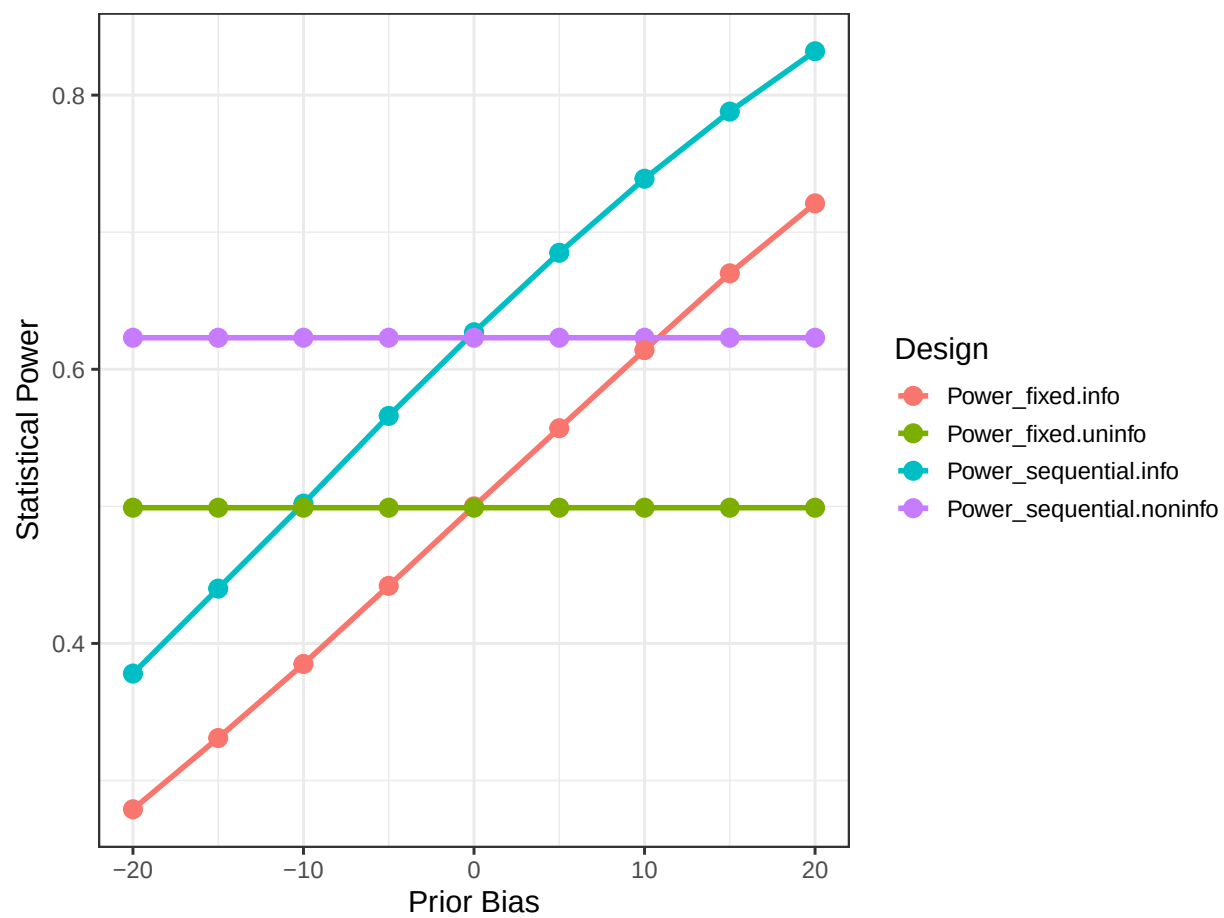


Figure 3: Power with varying prior bias

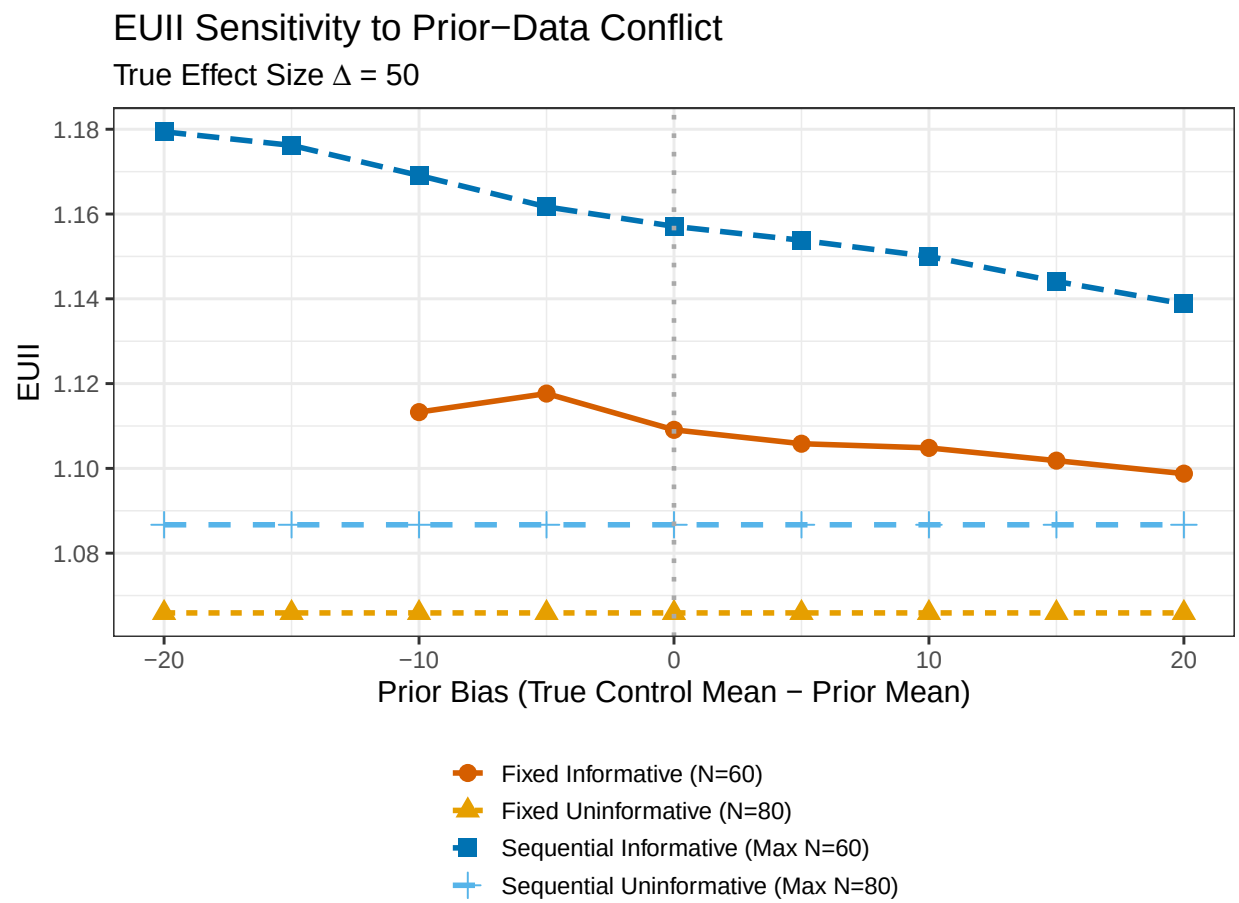


Figure 4: EUII with varying prior bias

3 Example 3.2: Oncology phase II single-arm trials

This example is a single-arm oncology phase II trial where we have to decide if a new treatment has sufficient effect to invest in further development. Simon two-stage design is usually applied in these cases. It is a *two stage design* which allows for early stopping for futility but **No early stopping for success**. The design aims at minimizing the number of patients treated with a drug with low activity.

To implement it in `gsbDesign`, we need normal approximation. It is well known that for binary data, the normal approximation is more appropriate on the logit scale. Let r_{Ei} be the number of responses and n_{Ei} the number of treated patients in stage i , $i = c, E$. If we denote y_{Ei} the log-odds of the observed response rate, its distribution is approximately:

$$y_{Ei} \sim N(\text{logit}(\pi_E), \sigma_E^2/n_{Ei})$$

where π_E is the underlying true response rate and $\sigma_E^2 = 1/(\pi_E) + 1/(1 - \pi_E)$.

We need to mimic Simon's two-stage design, so we introduce a virtual control arm with no patients, n_c with an extremely informative prior for π_c (prior centered at the true value with zero variance).

We set $\pi_c^* = 0.4$, $\pi_E^* = 0.6$, $\alpha = 0.05$, and $\beta = 0.20$. According to Simon's optimal two stage design, $n_{E1} = 16$ patients are recruited in the first stage and $n_{E2} = 30$ in the second stage. The trial is stopped for futility at the interim analysis if seven or less patients respond. The trial is a success at the end of the trial if more than 23 patients respond. Translating these criteria to the log-odds scale, the futility criterion is

$$p(\delta < \text{logit}(7.5/16) - \text{logit}(0.4) \mid \text{Data}) \geq 0.5$$

and the success criterion is

$$p(\delta > \text{logit}(23.5/46) - \text{logit}(0.4) \mid \text{Data}) \geq 0.5$$

where $\delta = \text{logit}(\pi_E) - \text{logit}(\pi_C)$.

Note that early stopping for success is not allowed, this means that $\mathbb{E}[N_+] = 46$ always.

3.1 Results

Table 2: Expected Total Sample Sizes ($\mathbb{E}[N]$) for the Simon Two-Stage Optimal Design.

control	treatment	delta	stage1	stage2
0.40	0.40	0.00	16.00	24.70
0.40	0.45	0.05	16.00	29.23
0.40	0.50	0.10	16.00	33.92
0.40	0.55	0.15	16.00	38.21
0.40	0.60	0.20	16.00	41.57

Table 3: EUII and other operating characteristics

True Rate delta ($\pi_E - \pi_C$)	DOR	EUII	$\mathbb{E}[N_+]$	$\mathbb{E}[N_-]$
0.05	3.60	1.0321	46.0	23.8
0.10	10.42	1.0648	46.0	23.8
0.15	27.22	1.1025	46.0	23.7
0.20	70.85	1.1490	46.0	23.6

Interestingly, the value of the prior for the alternative, doesn't have much influence here.

Operating Characteristics

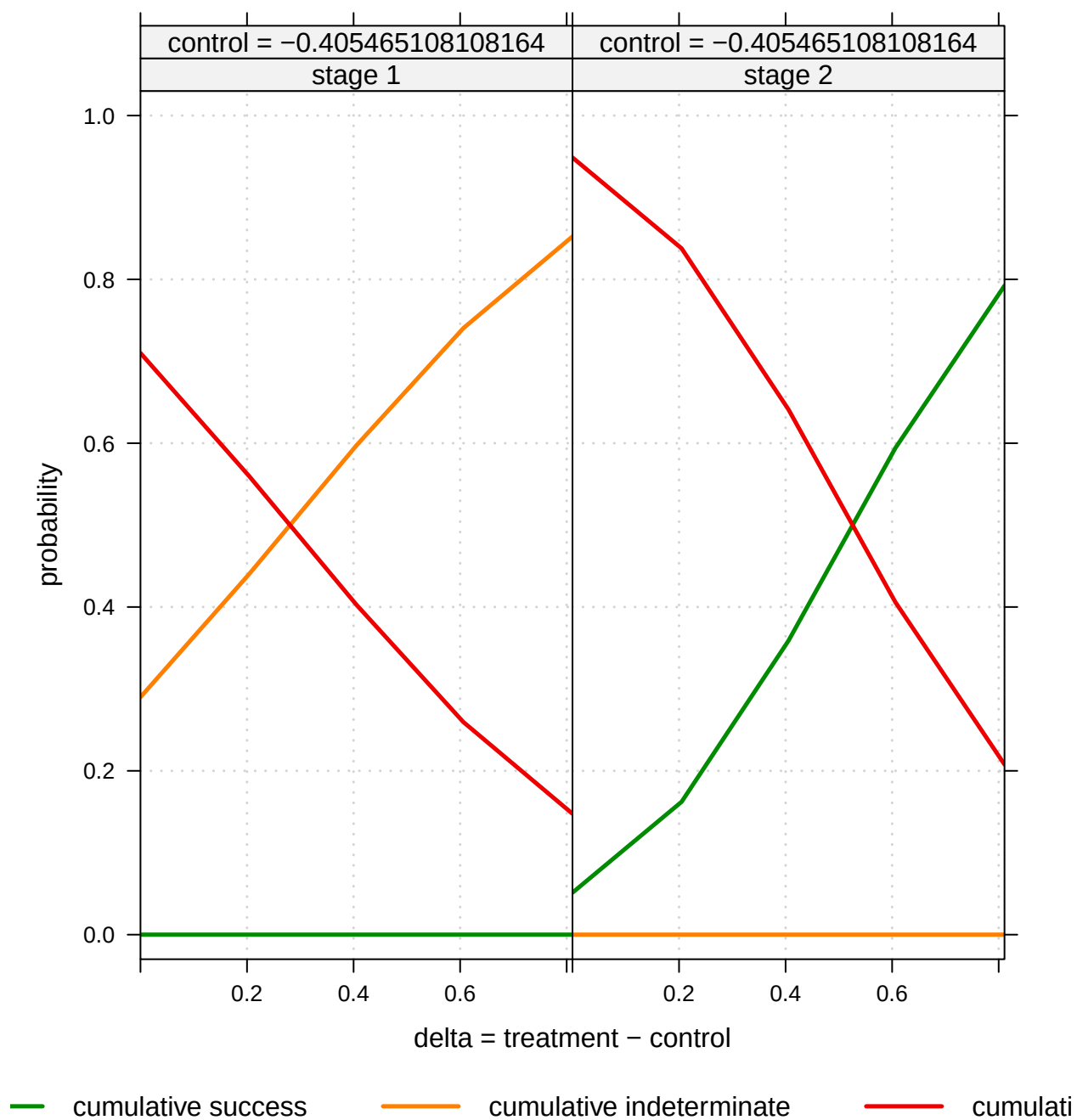


Figure 5: Operating Characteristics.

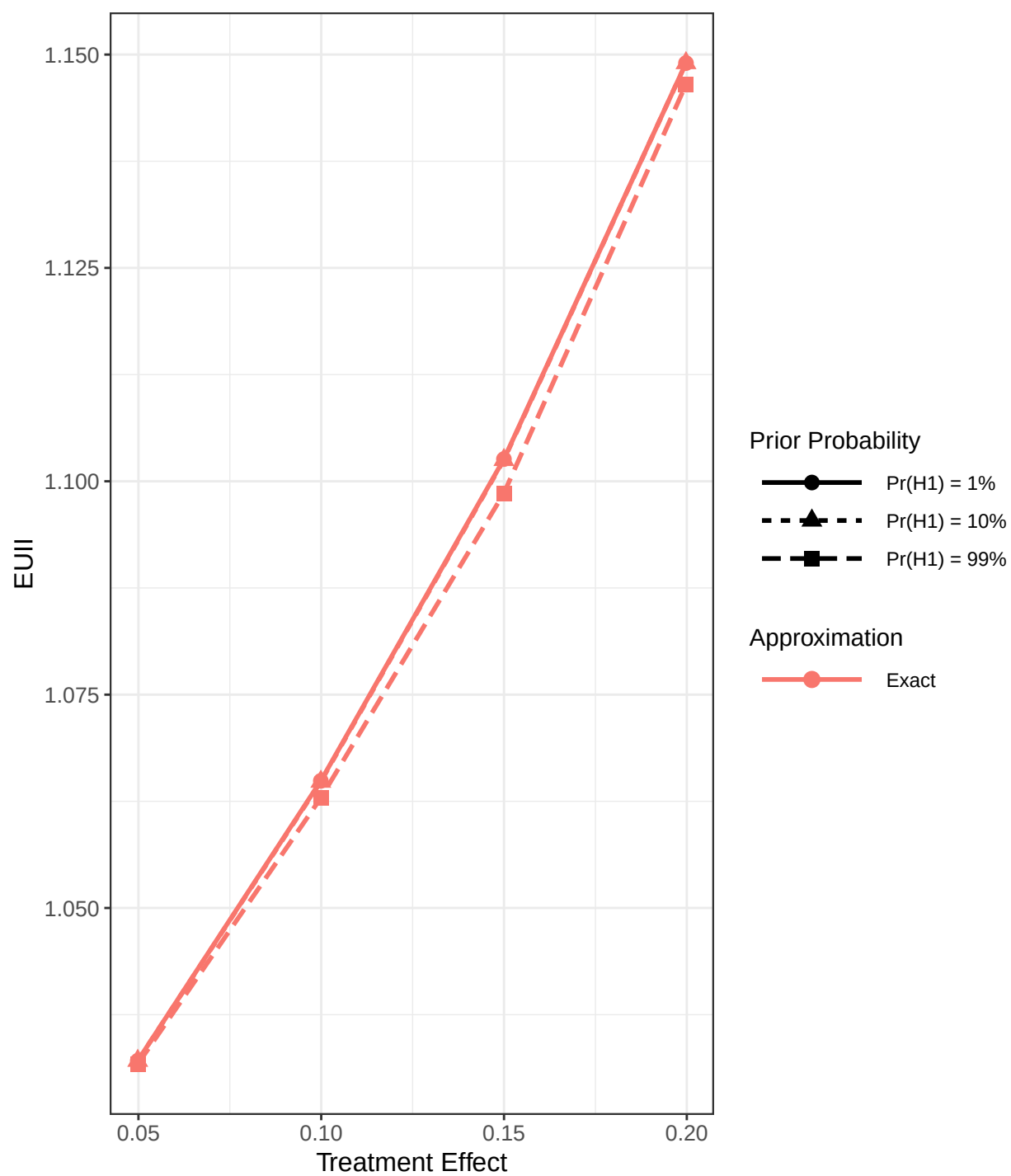


Figure 6: EUII with varying priors

4 Example 3.3. Design of a phase II trial in multiple sclerosis with count endpoint

In this example, the endpoint is MRI lesions **count**. The authors propose a Negative Binomial regression to model the count. Typically, this model is fit using a log link function. We call κ the assumed common over-dispersion parameter and μ_c and μ_E the underlying true mean MRI counts for control and experimental unit groups. Then, using the estimates r_c and r_E . We can write the normal approximation as:

$$\log(r_j) \sim N(\log(\mu_j), 1/\mu_j n_j + \kappa/n_j)$$

So we can define $\sigma_j^2 = 1/\mu_j + \kappa$. This comes from the use of delta method following the parameterization of a negative binomial distribution given by [2]:

$$f(y) = \frac{\Gamma(y+1/k)(\kappa\mu)^y}{\Gamma(y+1)\Gamma(1/k)(1+\kappa\mu)^{y+1/k}} \quad \text{for } y = 0, 1, 2, \dots$$

$$\text{Dispersion} = \kappa \quad \text{Var}(y) = \mu + \kappa\mu^2 \quad \mathbb{E}[y] = \mu$$

Variance computation with Delta Method. We know the delta method approximates the variance of a function of a random variable as:

$$\text{Var}[g(\hat{\theta})] \approx [g'(\theta)]^2 \text{Var}(\hat{\theta})$$

For the log transformation $g(\mu) = \log(\mu)$ where $g'(\mu) = 1/\mu$:

$$\text{Var}[\log(\hat{\mu})] \approx \frac{\text{Var}(\hat{\mu})}{\mu^2}$$

Using the negative binomial variance $\text{Var}(\hat{\mu}) = \mu + \kappa\mu^2$:

$$\text{Var}[\log(\hat{\mu})] \approx \frac{\mu + \kappa\mu^2}{\mu^2} = \frac{1}{\mu} + \kappa$$

□

Note that the variance of the log rate depends both on the mean μ and the over-dispersion parameter κ .

The design is a two-arms with 100 total subjects randomized with a 1:1 ratio and the interim analysis is planned when final outcome is observed for 50% patients, this mean that the interim analysis is planned when 25 patients per arm have completed the study.

Following Gsponer[1], we fix $\kappa = 2$ and $\mu_c = 14.8$.

Denote $\delta = \log(\mu_c/\mu_E)$, to stop for early success, the following double criterion must be met:

$$P(\delta > 0 \mid \text{Data}) > 0.9 \quad \text{and} \quad P(\delta > \log(2.5) \mid \text{Data}) > 0.5.$$

This is equivalent to having a posterior median for δ corresponding to at least a 60% relative reduction, with a 90% posterior probability of some reduction.

Whereas, in the case of stopping early for futility, a recommendation to stop would be based on the following double criterion:

$$P(\delta < \log(1.43) \mid \text{Data}) > 0.5 \quad \text{and} \quad P(\delta < \log(2.5) \mid \text{Data}) > 0.9.$$

This rule is equivalent to having a posterior median for δ corresponding to a risk reduction of less than 30%, and a 90% posterior probability of δ having a risk reduction of less than 60%.

Additionally, an enthusiastic prior is obtained from Phase II trial, and it is not placed on the placebo, like the previous design, but **on the treatment effect** δ . They observed a risk reduction of 75% (i.e. $\delta = \log(100/25) = \log(4)$) in a trial consisting of 80 patients per group, with placebo rate and over-dispersion parameter equal to the planning assumptions stated previously. This prior is considered to be worth 10 patients per group.

4.1 Results

Table 4 illustrates the Expected Total Sample Size ($E[N]$) across varying true treatment effects for both the Non-Informative and Informative. Since the trial needs at least 50 patients, values close to 50 indicate a high probability of early stopping, while values close to 100 indicates that the trial frequently required both stages to reach a decision.

Intuitively, when the true treatment effect is close to the prior’s value (e.g., 75% relative reduction), the Informative design proves to require less patients.

However, if the drug underperforms with respect to the prior, the sample size of the informative increases. At a 0% relative reduction (the null hypothesis), the Non-Informative design identifies futility efficiently, stopping the trial at an expected 59.1 patients. In contrast, the Informative design requires an expected 75.8 patients.

Table 4: Comparison of Expected Total Sample Sizes ($E[N]$) under Non-Informative and Informative Priors.

True Reduction (%)	$E[N]$ Non-Informative	$E[N]$ Informative
0	58.9	75.8
5	60.4	77.7
10	62.2	79.5
15	64.2	81.0
20	66.3	82.2
25	68.5	82.9
30	70.7	83.1
35	72.7	82.5
40	74.3	80.9
45	75.1	78.5
50	75.0	75.0
55	73.6	70.7
60	70.6	65.7
65	66.3	60.7
70	61.2	56.2
75	56.1	52.8
80	52.4	50.9
85	50.5	50.1
90	50.0	50.0

Figure 7 shows that the EUII is strictly larger for the Non-Informative design across every single scenario. The reason can be come from that while the informative prior saves about 3 to 4 patients when correctly rejecting H_0 , it costs many patients when failing to reject it. This mean that the EUII is dragged down.

Another reason can be that having the enthusiastic prior leads to an increased Power for some alternatives, but also increased T1E. This can in turn lead to a reduced DOR.

Note, when the true reduction is 85% and 90%, $P(-) = 0$, this mean that it is impossible to compute the EUII since the trial always ends up in a success.

Table 5: Comparison of EUH and Conditional Expected Sample Sizes between informative and non informative

True Red. (%)	EUH (Non-Inf)	EUH (Inf)	$E[N_+]$ (Non-Inf)	$E[N_+]$ (Inf)	$E[N_-]$ (Non-Inf)	$E[N_-]$ (Inf)
5	1.0071	1.0058	50.0	50.1	59.2	77.0
10	1.0141	1.0117	50.4	50.3	59.4	77.2
15	1.0213	1.0175	50.7	50.4	59.6	77.5
20	1.0281	1.0235	51.0	50.4	59.8	77.7
25	1.0356	1.0297	51.4	50.8	60.0	77.8
30	1.0427	1.0359	52.2	51.2	60.2	77.9
35	1.0501	1.0424	53.3	51.8	60.5	78.0
40	1.0576	1.0490	54.4	52.4	60.6	78.0
45	1.0652	1.0560	55.8	53.1	60.7	77.9
50	1.0734	1.0635	57.0	53.7	60.7	77.8
55	1.0827	1.0720	57.9	54.1	60.5	77.6
60	1.0937	1.0820	58.3	54.2	60.2	77.3
65	1.1078	1.0946	57.9	53.8	59.8	77.1
70	1.1268	1.1112	56.5	52.8	59.4	76.9
75	1.1539	1.1337	54.3	51.6	59.1	76.8
80	1.1933	1.1687	51.9	50.6	59.0	76.8
85			50.5	50.1		
90			50.0	50.0		

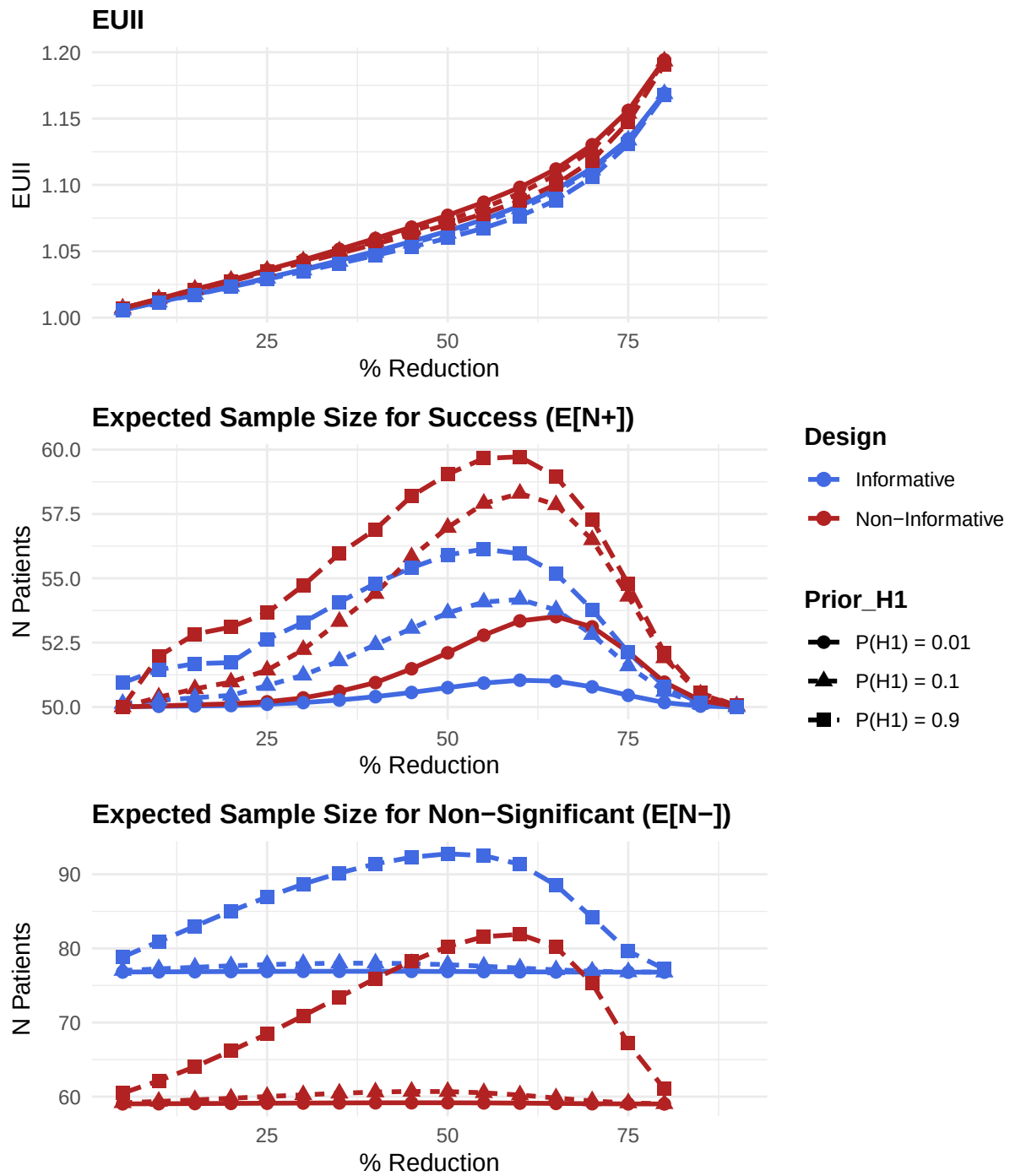


Figure 7: EUII under varying priors.

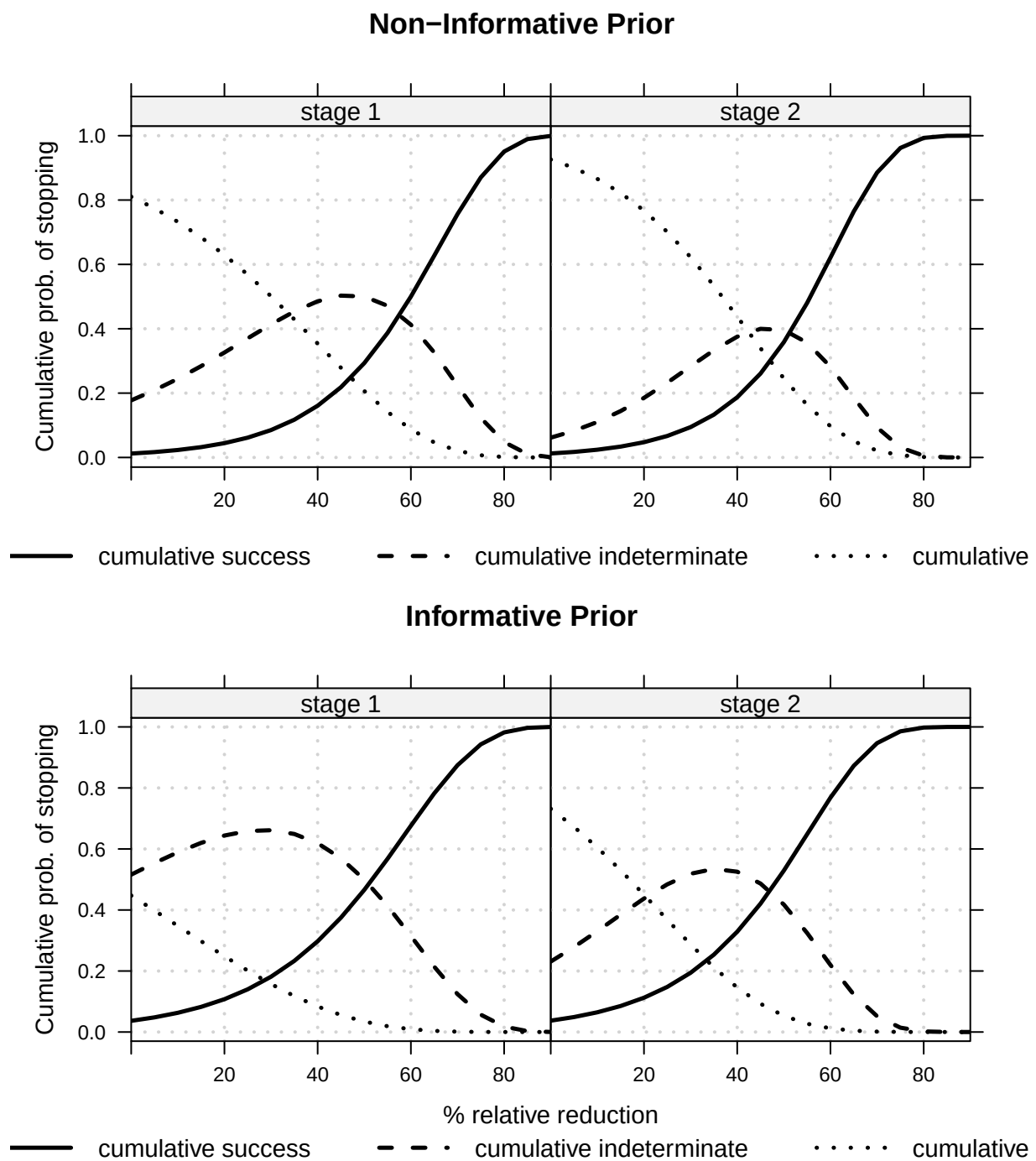


Figure 8: Cumulative probability of stopping for the Non-Informative and Informative prior designs.

5 Example 3.4. A phase III trial design with Bayesian futility criteria for a time-to-event endpoint

In the context of survival analysis, the measure of treatment effect used is often Hazard Ratio.

Let $\lambda_1(t)$ and $\lambda_0(t)$ be the hazard functions for the test and control groups, respectively. Under the proportional hazards assumption, the hazard ratio is constant over time: $HR = \lambda_1(t)/\lambda_0(t)$.

We define the treatment effect $\theta = \log(HR)$ and we can use the standard asymptotic normal approximation for the estimate of the log hazard ratio under the assumption of equal survival distributions:

$$\hat{\theta} \sim N\left(\theta, \frac{c}{E_1 + E_2}\right)$$

for a k:1 randomization where $c = (k+1)^2/k$ and E_1 E_2 are the events in test and control group. For a 1:1 randomization, $c = 4$ $E_1 + E_2 = E$ and we can define:

$$\text{Var}(\hat{\theta}) = \frac{4}{E}$$

Since gsbDesign computes the variance of $\hat{\delta}$ as

$$\text{Var}(\hat{\delta}) = \frac{\sigma_E^2}{n_E} + \frac{\sigma_C^2}{n_C}$$

Setting $\sigma_T = 1$ and $\sigma_C = 1$,

$$\text{Var}(\hat{\delta}) = \frac{1^2}{E_{arm}} + \frac{1^2}{E_{arm}} = \frac{2}{E_{arm}}$$

Since the total number of events across both arms is $E_{total} = 2 \times E_{arm}$, we can substitute $E_{arm} = E_{total}/2$:

$$\text{Var}(\hat{\delta}) = \frac{2}{E_{total}/2} = \frac{4}{E_{total}}$$

. So by setting `sigma = c(1,1)` in the paper we mapped the asymptotic variance into the gsbDDesign.

For a more general case using hazard ratios, the authors suggest to use the formula $\sigma_1 = \sigma_2 = \sqrt{cE_1E_2/(E_1 + E_2)^2}$ to implement in the software. To represent positive values as beneficial treatment effect, we define the parameter $\delta = -\theta = \log(1/HR)$. Consequently, a hazard ratio less than 1 (indicating a reduction in the risk of an event) results in a positive value for δ . The non-inferiority margin is defined as $HR_{NI} = 1.075$, which corresponds to $\delta_{NI} = -\log(1.075) \approx -0.0723$.

The design consists of three stages with interim analyses after 1/3 and 2/3 of the total expected events have occurred. Assuming $E_{total} = 1800$ events, each stage accrues 600 events. Under the null hypothesis of equal survival distributions ($E_1 = E_2$), this corresponds to 300 events per treatment group per stage.

The original trial used modified Haybittle Peto boundaries for stopping for success, requiring a high posterior probability of superiority at the interims ($P(\delta > 0|D) > 0.99995$ and $P(\delta > 0|D) > 0.9995$) and a standard probability of non-inferiority at the final analysis ($P(\delta > \delta_{NI}|D) > 0.975$).

The final trial was a massive waste of money since it concluded without any evidence. The paper argue that adding bayesian futility stopping rules could have saved resources. Stage 1: Stop if $P(\delta < \delta_{NI}|D) > 0.95$ Stage 2: Stop if $P(\delta < \delta_{NI}|D) > 0.90$ Stage 3: Stop if $P(\delta < \delta_{NI}|D) > 0.80$

5.1 Results

True HR	EUII (Orig)	EUII (Bayes)	$E[Events_+]$ (Orig)	$E[Events_+]$ (Bayes)	$E[Events_-]$ (Orig)	$E[Events_-]$ (Bayes)
0.861	1.0056	1.0060	1669.394	1669.394	1786.466	1694.805
0.900	1.0040	1.0042	1761.648	1761.648	1786.517	1695.199
0.951	1.0026	1.0027	1794.387	1794.387	1786.855	1697.699
1.000	1.0017	1.0017	1798.929	1798.929	1787.352	1701.165
1.051	1.0006	1.0006	1800.000	1800.000	1787.204	1699.299
1.073	1.0001	1.0001	1800.000	1800.000	1786.580	1695.414
1.100	0.9993	0.9993	1800.000	1800.000	1784.521	1687.101
1.150					1774.957	1663.120

Note the EUII smaller than 1 when HR larger than 1.075 (since it is a noninferiority trial). This because if HR larger than 1, the drug is actually harmful. This lead to a $LR_+ < 1$, so $Power < T1E$ and as established by Lehmann (1959), a test where the power to reject the null is smaller than the T1E rate is formally called a "biased test".

This mean that DOR is smaller than 1, which in turn lead to a EUII smaller 1. Recall that $DOR = O(H1|+)/O(H1|-)$, where $O(H1|+)$ is the odds of H1 being true given a significant result. If $DOR < 1$, then a significant result leaves you with lower odds of H1 being true than a non-significant result.

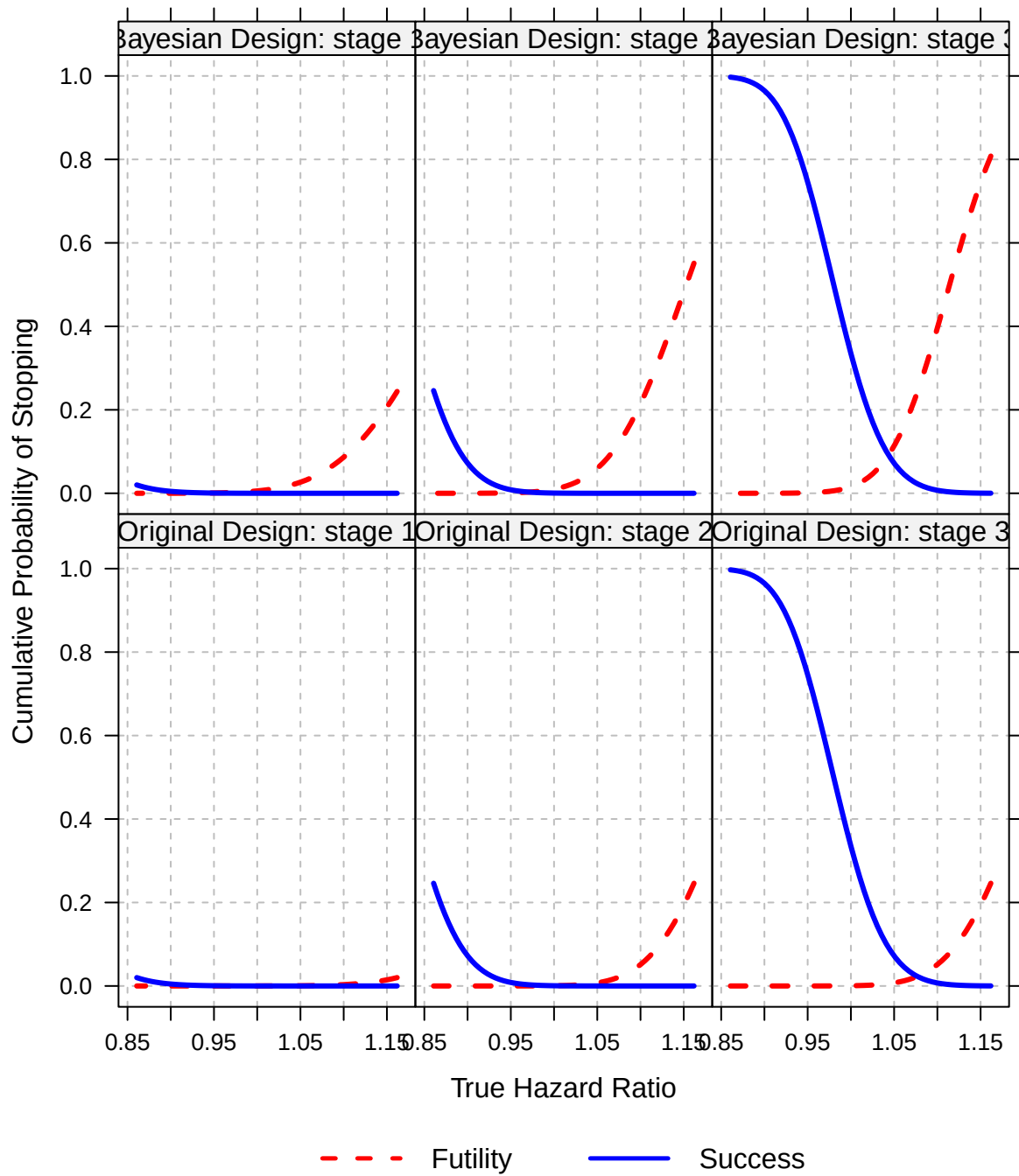


Figure 9: Hazard Ratio

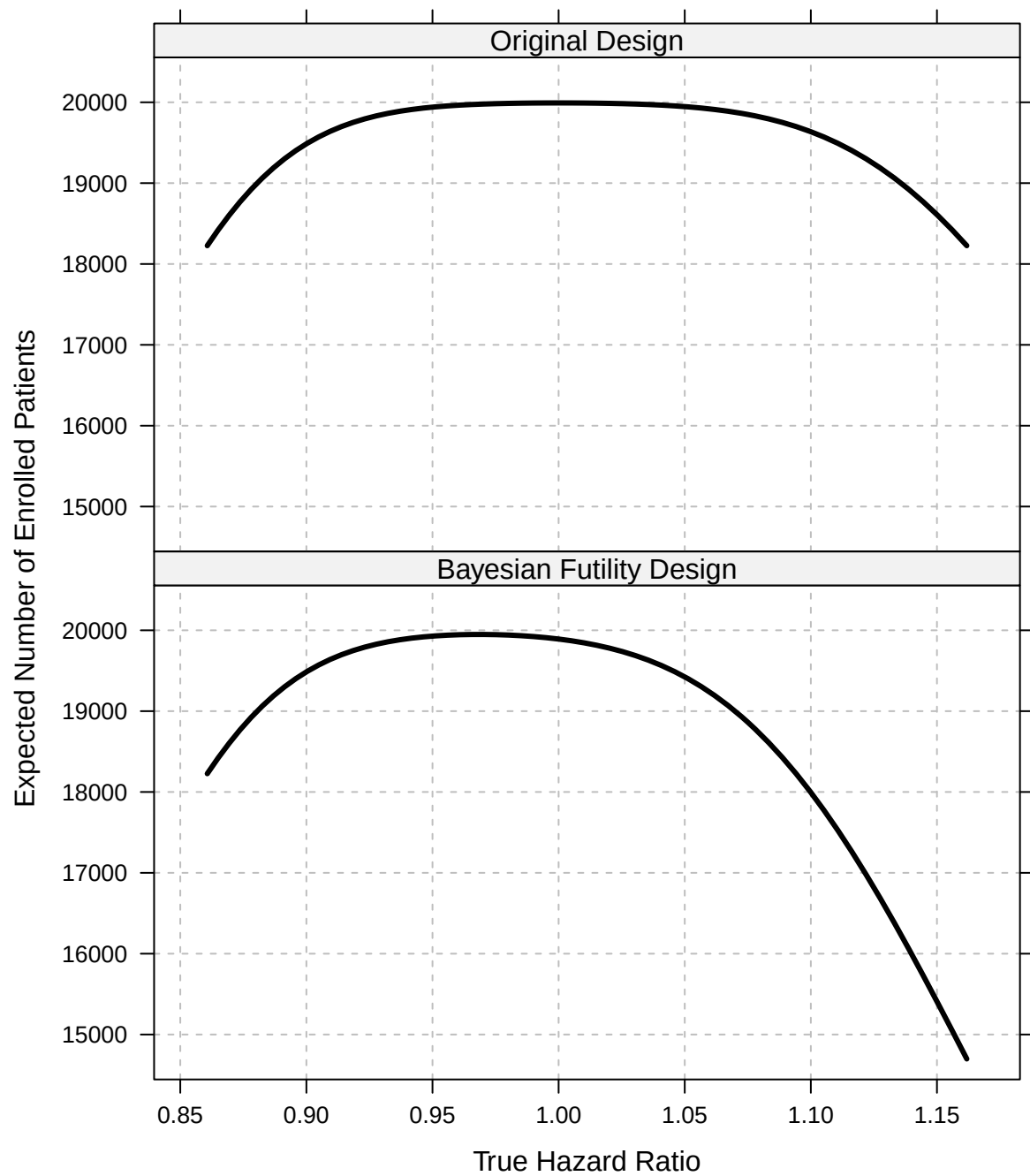


Figure 10: Hazard Ratio

References

- [1] Thomas Gsponer, Florian Gerber, Björn Bornkamp, David Ohlssen, Marc Vandemeulebroecke, and Heinz Schmidli. A practical guide to Bayesian group sequential designs. *Pharmaceutical Statistics*, 13(1):71–80, 2014. doi: 10.1002/pst.1593.
- [2] Oliver N. Keene, Mark R. K. Jones, Peter W. Lane, and Julie Anderson. Analysis of exacerbation rates in asthma and chronic obstructive pulmonary disease: example from the tristan study. *Pharmaceutical Statistics*, 6(2):89–97, 2007. doi: 10.1002/pst.250.